



Module	SE101.3 -OOP with Java – 25.3
Lab sheet Number	04
Areas covered	Methods and Constructors

1. What is a Constructor in Java? Explain with an example.
2. Create a Java class to calculate the addition of two integer values using a function for Addition.
3. What is a Method in Java? Explain with an example.
4. Create a Java class named RectangleAreaCalculator that calculates the area of a rectangle.

The class should have:

- Private instance variables length and width to store the dimensions of the rectangle.
- A constructor initializes the length and width variables.
- A method named calculateArea () returns the area of the rectangle by multiplying length and width.

In the main method:

- Create an instance of RectangleAreaCalculator with length set to the value you prefer, and width set to the value you prefer.

- Calculate and display the area of the rectangle.

5. Create a Java class named Car with the following specifications:

- It should have instance variables make, model, and year to store the make, model, and year of the car, respectively.
- Provide a constructor that initializes these instance variables.
- Create a method named displayDetails () to display the make, model, and year of the car.

6. Create a Java class named SecurityAlertSystem.

The class should contain:

- alertID
- threatType
- severityLevel

Requirements:

- a) Create a parameterized constructor to initialize all values.
- b) Create a method named displayAlert() to display the alert details.
- c) In the main method, create two security alerts and display their information